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PI Name:	Lipa, John A. Ph.D.		
Project Title:	Fundamental Physics Experiments With Superconducting Cavity - Stabilized Oscillators on Space Station		
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PI Email:	John.Lipa@stanford.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	650-723-4562
Organization Name:	Stanford University		
PI Address 1:	Physics Department		
PI Address 2:	Stanford University		
PI Web Page:			
City:	Stanford	**State:**	CA
Zip Code:	94305	**Congressional District:**	14
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Contact Monitor:		**Contact Phone:**	
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Task Description:	The superconducting cavity-stabilized oscillator project was developed to perform an improved search for Lorentz transformation violations in Special Relativity in space. Small but non-zero violations have been suggested as occurring due to new physical phenomena at extremely high energies, such as encountered in the Big Bang. The experiment would be one of the few ways of probing such physics. We developed a new generation of microwave cavities that are much less sensitive to acceleration than earlier models. This should allow us to reach stability levels in the 10^-17 range if other conditions are met. These conditions are improved temperature control and a lower noise phase-locked loop for the cavity. Work was started on the systems to meet these requirements. This stability level represents the threshold where new physical phenomena might be encountered. The superconducting microwave oscillators are intended for deployment on the International Space Station within the Low Temperature Microgravity Physics Facility being		

	developed by JPL. Signals from the apparatus could also be used by the Primary Atomic Reference Clock experiment for conducting General Relativity experiments on Space Station. The technology developed for the experiment can be used in other applications where high quality reference oscillators are needed, for example for precision Doppler tracking using the Deep Space network.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The technology developed for the experiment can be used in other applications where high quality reference oscillators are needed, for example for precision Doppler tracking using the Deep Space network. The project also developed improved electronics for locking slave oscillators to high-Q resonators and the method was applied to quartz crystal oscillators that are used in many precision timing applications, both in space and on earth. State-of-the-art results were obtained.
Task Progress:	The superconducting cavity-stabilized oscillator project was developed to perform an improved search for Lorentz transformation violations in Special Relativity in space. Small but non-zero violations have been suggested as occurring due to new physical phenomena at extremely high energies, such as encountered in the Big Bang. It was to use the Low Temperature Microgravity Physics Facility (LTMPF) facility that was being developed by JPL for deployment on the International Space Station (ISS). Unfortunately the flight portion of this program was cancelled in 2001, along with the entire suite of fundamental physics experiments slated for the ISS. Since some possibility still exists for a re-start or a free flier mission, we continued work on the project focusing on Lorentz violation tests on the ground and on better defining the flight measurements. We also worked on developing a special cryocooler for use in the LTMPF and on an optical version of the experiment, which has the potential for providing much more stringent tests of relativity. The LPI principle of GR states that the outcome of any non-gravitational experiment conducted in a local, free-falling frame is independent of where and when that experiment is conducted. A consequence is that different types of clocks keep exactly the same time, no matter where they are co-located in the universe, provided that the laws of physics do not vary from place to place. One of our goals is to look for a violation of LPI by comparing a microwave cavity frequency with that of an atomic clock as a function of position and gravitational potential as the ISS orbits the earth and the sun. A microwave cavity and an atomic clock have different dependencies on fundamental physical constants that can potentially reveal a violation of LPE. Alternatively, one can view the experiment as setting limits on effects predicted by various theories competing with general relativity as descriptions of the interaction of matter and spacetime. This test could improve on the earlier measurement by a factor of about 1000. The LTMPF, designed for the ISS, would allow relatively easy access to temperatures as low as 1.5 K in space for a wide range of missions with durations of up to a few months. Also, a variety of atomic clocks are being developed for flight on the ISS, with very good stability for time periods in excess of 1,000 seconds. In collaboration with the Jet Propulsion Laboratory we developed an improved superconducting microwave oscillator (SUMO) which could significantly augment the scope and capability of the space clock ensemble. SUMO can provide a high stability, low phase noise signal capable of being slaved to atomic clocks, providing them with a ‘flywheel’ and greatly enhancing their short term performance. If linked to an atomic clock it would provide a low noise reference oscillator and allow differential frequency measurements against an atomic standard. SUMO is designed to be an insert in the LTMPF and would require three to six months of operation in order to meet all its science objectives. Recently Kostelecky and Mews (KM) pointed out that in the Standard Model Extension (SME), which describes general Lorentz violations, a number of terms exist which may lead to detectable effects in electromagnetic cavity experiments. In general the fractional beat frequency between two resonators with non-spherical mode symmetry is expected to be given by a sinusoidal function involving the SME parameters and the rotation rate and velocity of the cavities relative to inertial space, typically taken as the cosmic microwave background frame. A typical experiment that studies this beat note is then able to put bounds on the various parameters using standard statistical techniques. The coefficients in the function include effects that probe for additional physics possibly due to residual phenomena left over from the birth of the universe. Due to the extreme sensitivity of modern clocks it is possible to put useful bounds on such possibilities. As pointed out by Kostelecky and Potting, in some circumstances Lorentz violations originating at the Plank scale may manifest themselves at a fractional frequency variation at the 10^-17 level in the absence of suppression factors. Thus measurements in this range or below could result in some interesting new physics that goes beyond the Standard Model and would of course have implications for cosmology. There are nineteen possible Lorentz-violating parameters contributing to the coefficients, of which only ten were bounded when the experiment was started. The present experiment is sensitive to seven linear combinations of the remaining nine parameters. In principle, bounds can be set on the individual parameters using numerical techniques. Of course, ground measurements can be made to determine the parameters, but these are limited by the lower modulation velocities and difficulties with spurious accelerations. Briefly, the apparatus developed for the project consists of a pair of cylindrical superconducting microwave resonant cavities formed from highly annealed niobium and operating with an internal ultrahigh vacuum at about 1.2 K. High vacuum conditions for the cavities are maintained by means of a permanent internal vacuum with a pinch off seal. High vacuum conditions are also needed externally to reduce the pressure sensitivity of the devices. The cavities operate in the TM010 mode that gives sensitivity to the velocity of light in radial directions relative to the axis of symmetry. A Cryoperm magnetic shield is used to reduce the field at the cavity to ~ 3x10^-3 Gauss which is necessary to maintain the cavity Q on the order of 10^10. The connections between the cavities and the room temperature electronics are made using semi-rigid stainless steel coaxial cables. The electronics includes a pair of conventional microwave oscillators with frequencies locked to the cavity resonances. A Pound stabilization scheme is used on each cavity to attenuate the effect of phase shifts in the cables. The outputs of the oscillators are beat together and then with an r.f. signal to produce a low audio frequency difference signal. This technique allows one to track the relative phase of two microwave signals to the milli-radian level by accurately timing the zero crossings of the beat note. Accelerations of a cavity, such as that due to the earth's gravitational pull, cause distortions in its shape. The component of these distortions that is perpendicular to the cylindrical cavity wall causes a shift in the cavity resonant frequency. To minimize the frequency changes that occur due to changing accelerations we have designed a supporting structure for the cavity that takes advantage of some of the symmetries in its design. Supporting the cavity radially around its center reduces the sensitivity of the frequency to axial acceleration because of symmetry. With an end-support the calculated frequency sensitivity delta upsilon/upsilon is about 6.5x10^-9 per g. Allowing for machining errors, a realistic center-support design gives a sensitivity of ~ 10^-12. By varying the mass attached to the ends of the cavity one can tune the frequency shift to an arbitrarily small value. To date we have been able to reduce the sensitivity to delta

upsilon/upsilon is < 7x10^-11 per g. On the ISS the residual accelerations at very low frequencies are mostly caused by aerodynamic drag and are estimated to be below 10^-6 g, potentially allowing frequency discrimination to the 10^-17 range. During testing we found that the cavities exhibit some transverse acceleration sensitivity. Computer modeling indicates that this coefficient should be zero if there is perfect axial symmetry in the cavity and support structure. However, in a practical device, this symmetry is broken to some degree and a non-zero coefficient results. To correct for this effect the support system needs to be slightly modified with a compensating asymmetry. We began investigating weak spring supports of various designs to develop transverse nulling strategies. We were able to modify the coefficient and developed a support structure that is more amenable to fine-tuning of radial asymmetries. With this system we demonstrated tuning to the 10% level.

Near 1.3K the temperature coefficient of the cavity frequency is typically delta upsilon/upsilon ~ 3x10^-10 per degree. Thus to achieve a signal stability of 10^-17 at orbital periods we need to control the cavity temperature to about ± 3x10^-8 K. The flight of the CHEX experiment on the Shuttle demonstrated that this is within the range of existing capability using high resolution thermometry techniques. The thermometry used on CHEX was a specially developed paramagnetic salt detector with a superconducting magnetometer readout system. It was found possible to obtain an rms short-term thermal stability below 3x10^-8 K, comparable with our needs. On time scales of minutes the rms stability was < 2x10^-9 K exeeding our requirement. We are now using a modified version of this thermometry technique to control the temperature of the cavity, minimizing temperature-induced frequency variations and have achieved a thermal stability of ± 2x10^-8 K over periods of hours.

Recently we operated two cavities aligned to detect signals of the form given in the above equation using the earth's rotation as the source of modulation. Due to the demands of the test program for the flight experiment, only occasional records are available with more than 24 hours of uninterrupted data. Our analysis of nine sets obtained over an 80 day period set a limit of <3x10^-13 on the amplitude of any signal. At this level the signal stability appeared to be limited by the transverse sensitivity of the cavity which was mounted vertically. This measurement put the first bounds of < 10^-12 on four sets of parameters in the SME theory, and < 10^-8 on three others. Additional measurements could be performed with the transverse coefficient more accurately nulled. Our results were published in Physical Review Letters in 2003. As part of the effort to improve the performance of the superconducting cavities we performed some experiments to find a viable way to operate below the ambient temperature of the LTMPF cryogenic environment. We developed an improved continuously pumped cooler that was capable of operating slightly below 1 K using a small amount of the facility cryogen. The cooler was developed to address the potential need for temperatures of < 1.2 K where the Q of niobium cavities can exceed 10^11. The LTMPF cryostat was designed for TB ~1.6 K prior to the selection of the present experiment for flight development, but it could be pushed to 1.5 K. We found that that the application of heat to the fill line of a conventional cryocooler dramatically enhances its cooling capability. For TB = 1.6 K, we achieved Tcooler < 0.95 K. The operation of the refrigerator in this mode lasted more than 24 hours and the base temperature was stable. For Tcooler = 1.2 K, the refrigerator provided a cooling power of about 100 µW which would be adequate for the present experiment. The addition of a cooler capable of reaching 0.95 K within a superfluid bath was found to be a lower cost alternative than a redesign of the dewar exhaust system to reach such low temperatures.

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